Dr.B. Ravi, Assistant Professor, Department of ECE, IIIT Trichy

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To strive for attaining academic and research excellence through continuous learning

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Web of Science ResearcherID:

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Google Scholar Link:

https://scholar.google.co.in/citations?user=Y6WmauoAAAAJhl=enoi=ao

Scopus ID:

https://www.scopus.com/authid/detail.uri?authorId=57197836657

ResearchGate Link:

https://www.researchgate.net/profile/Banoth-Ravi

Publons Link:

https://publons.com/researcher/3190214/banoth-ravi/

Personal Webpage Link:

1. http://banothravi.byethost6.com/index.php; 2. https://sites.google.com/view/ravibanothiitism

Research Interests:

Machine Learning for Wireless Communication and Networks, 5G/6G Beyond Networks, Data Communications and Computer Networking, Computer and Communication Networks, Wireless Ad Hoc Networks, Knowledge Defined Networking, Vehicular Communications, Intelligent Transportation Systems, Intelligent Digital Twin Networks, Software Defined Networks, Vehicular Networks, Queueing Theory, Data Networks, Stochastic Networks, Optimization Techniques, Computer Systems Performance Evaluation and Prediction, and Quantum Communication and Networks.

Anusandhan National Research Foundation (ANRF) – Proposals Submitted

- Stochastic Digital Twin-Driven Queue Optimization for Zero-Touch Ultra-Reliable Communication in 6G Software-Defined Vehicular Networks (SDVNs)-Inclusivity Research Grant: ANRF/IRG/2025/001112/ENS, 31-Jul-2025, Not Recommended.
- Optimizing Dynamic Spectrum Allocation and Queue Prioritization for Real-Time Healthcare in 5G and Beyond Networks:Prime Minister Early Career Research Grant: ANRF/ECRG/2024/005535/ENS, 19-Nov-2024, Not Recommended.
- Stochastic Digital Twin for Queue-Aware Zero-Touch Provisioning in Software-Defined Vehicular Networks:ARG-MATRICS: ANRF/ARGM/2025/003004/TS, 15-Jul-2025, Not Recommended.
- Development of a QoS-Optimized Queueing Framework for Real-Time Healthcare in 5G and Vehicle-to-Everything: Inclusivity Research Grant, ANRF/IRG/2024/000795/ENS, 10-Dec-2024, Not Recommended.
- A Stochastic Information-Theoretic Framework for Zero-Touch Route Optimization and Congestion Control in Digital Twin-Enabled 6G-SDVNs: PMRF, ANRF/IRG/2024/000795/ENS, 01-Dec-2025, Not Recommended.

Reviewed Journals and Conferences:

IEEE ANTS Conference, IEEE Transactions On Intelligent Vehicles, IEEE Transactions On Vehicular Technology, IEEE Transactions On Signal And Information Processing Over Networks, IEEE Internet Of Things Journal , IEEE Transactions On Consumer Electronics, IEEE Open Journal Of Vehicular Technology, IEEE Access, Springer Telecommunication Systems, Springer Journal Of Wireless Networks, Wiley Journal Of Transactions On Emerging Telecommunications Technologies, Measurement: Sensors, WILEY- International Journal Of Network Management, Frontiers In Energy Research, Plos One, Wiley-Internet Technology Letters, IET Networks, Scientific Reports etc.

Education:

- 15th-February-2016 – 09th-May 2022-Awarded: **Electronics Engineering**, *(Ph.D) Doctor of Philosophy*, **IIT(ISM), Dhanbad**.
  Working on Stochastic Analysis Vehicular Ad hoc Networks.
  Supervisor: Prof. Jaisingh Thangaraj
- October-2009–October-2011: **First Class with Distinction, Electronics and Communication Engineering**, *Master of Technology (Digital Systems & Computer Electronics)*, **JNTUH, Hyderabad**.
  Worked on Implementation of Multi-cast Routing Protocol for Wireless Ad-hoc Networks.
  Supervisor: Prof. Duggirala Srinivasa Rao
- August-2003–May-2007: **First Class, Electronics and Communication Engineering**, *Bachelor of Technology*, **JNTUH, Hyderabad**.
  Worked on GSM-based 3310 Smart Mobiles, which deals with receiving and making a call to remote areas. And we can send and receive messages from the mobile nodes.

Teaching Experience:

- 07^{th}-July-2007 to 30^{th}-07-2009 **Electronics and Communication Engineering**, *Assistant Professor*, DVR Institute of Technology, Hyderabad.
  Subjects Taught: Control Systems, Electronics Device & Circuits, Analog & Digital Communications, Digital Electronics, Communications Lab.
- 15^{th}-July-2011 to 14^{th}-May-2012 **Electronics and Communication Engineering**, *Assistant Professor*, DVR Institute of Technology, Hyderabad.
  Subjects Taught: Computer Networks, VLSI, IC Applications, Network Theory Lab.
- 02^{nd}-July-2012 to 03^{rd}-February-2016 **Electronics and Communication Engineering**, *Assistant Professor*, Mahaveer Institute of Science and Technology, Hyderabad.
  Subjects Taught: Data Communication & Computer Networks, Signals & Systems, VLSI, Electronics Instrumentation & Measurements, Computer Networks, Microprocessors, Antenna Theory & Wave Propagation, Coding Theory & Techniques, Advanced Data Communications.
- 15^{th}-February-2016 to 14^{th}-February-2021 **Electronics Engineering Department**, *Research Scholar cum Teaching Assistant at IIT(ISM) Dhanbad*.
  Teaching Assistant: Analog Circuits, Signals & Systems, Network Theory & Filter Design, Electronics Instrumentation & Measurements, Optical Communication, Basic Electronics Engineering, Computer Communication Networks

Instructors: Prof. Jaisingh Thangaraj and Prof. Jitendra Kuamr, EE Department, IIT (ISM) Dhanbad.

- 10^{th}-February-2023 to Till Now **Electronics and Communication Engineering Department**, *Assistant Professor at IIIT Tiruchirappalli*.
  Teaching: RF& Microwave Engineering, Probability and Random Process, Digital Principles and System Design, Digital Communication, Electronic Circuits, Communication Theory, Information Theory and Coding, Basics of Electronics Engineering, Digital System Design, Digital Principles and System Design Laboratory, Digital Communication Laboratory, Electronic Circuits Laboratory

Achievements/Scholarships:

- 2011 **FET**, *Qualified in Faculty Eligibility Test* , AndhraPradesh.
- 2008 **Qualified in GATE-2008**, *GATE, India*, Achieved 393 Score .
  93.33 Percentile.

Fellowship Award in M.Tech (2009 to 2011) from MHRD, Govt. of India.

- February 2016-March 2021 **Research Fellowship Award from MHRD, Govt. of India**, at Department of Electronics Engineering .
  IIT(ISM), Dhanbad, Jharkhand, India-826004.

Additional Responsibilities at IIIT Trichy:

- Deputy Warden (Boys) (i/c):2023-2024
- Student Discipline (Member):2025-2026
- Campus Maintenance (Member):2024-2025
- Mess Member:2024-2025
- National Service Scheme (NSS) Coordinator:2024-2025
- Class Coordinator
- Student Discipline (Member)
- Hostel Warden(i/c):2024-2025
- Chief Warden (i/c): 2025-till date
- Indoor Sports (In-charge): 2025–till date
- Hostel Furniture Procurement – Ongoing

- B.Tech Final Year Projects – Guided 21 Students
- Mess Tender Floated – AY 2023–2024 and 2024–2025

Personal Information:

Name: Banoth Ravi
Father's Name: Mangya
Date of Birth: 12^{th} August 1985
Marital Status: Married
Gender: Male
Nationality: Indian
Passport Details: S9347192, Date of Expiry:14/11/2028
Web of Science ResearcherID: AAD-3171-2019
Home Address: H.No.:14-64, Near Govt. High School, Seethampeta-PO, Garla-Mdl, Mahabubabad-Dist, Telangana-State, India-507210.

Web of Science (SCI)-Publications:

SCI-Journal (Published)

12. Banoth Ravi, Utkarsh Verma . "Spectrum Allocation in 5G and Beyond Intelligent Ubiquitous Networks", Accepted: November 6, 2024, International Journal of Network Management, DOI: 10.1002/nem.2315, (Science Citation Index (SCI): Journal: International Journal of Network Management (Wiley): IMPACT FACTOR: 1.5.

11. Chintala Sridhar, Murla Bhumi Reddy, Srihari Gude, Damodar Reddy Edla, and Banoth Ravi. "Least Mean Square/Fourth Adaptive algorithm for excision of ocular artifacts from EEG signals." Applied Acoustics 221 (2024): 110009. DOI: `https://doi.org/10.1016/j.apacoust.2024.110009`.

10. Bittu Kumar, Neeraj Kumar, Manoj Kumar, S. V. S. Prasad, Ashwini Kumar Varma, and Banoth Ravi. "Comparative Studies of Single-Channel Speech Enhancement Techniques." IETE Journal of Research (2023): 1-17, DOI: `https://doi.org/10.1080/03772063.2023.2273299`.

9. Manoj Kumar, Purnendu Shekhar Pandey, Banoth Ravi, Bittu Kumar, S. V. S. Prasad, Rajesh Singh, Santosh Kumar Choudhary, and Gyanendra Kumar Singh. "Impact of Sn-doping on the optoelectronic properties of zinc oxide crystal: DFT approach." Plos one 19, no. 1 (2024): e0296084, DOI: `https://doi.org/10.1371/journal.pone.0296084`.

8. Ravi, Banoth, et al. "Stochastic modeling for intelligent software-defined vehicular networks: A survey." Computers 12.8 (2023): 162, DOI: `https://doi.org/10.3390/computers12080162`.

7. Banoth Ravi "Stochastic Modeling and Performance Analysis in Balancing Load and Traffic for VANETs: A Review "***International Journal of Network Management (Q3),*** **(*Wiley*)**, DOI: `https://doi.org/10.1002/nem.2224` [**IF: 1.914**].

6. Banoth Ravi, Jaisingh Thangaraj, "Stochastic modelling and analysis of mobility models for intelligent software defined internet of vehicles, "***Physical Communication (Q3),*** **(*Elsevier*)**, DOI: `https://doi.org/10.1016/j.phycom.2021.101498` [**IF: 1.81**].

5. Banoth Ravi, Jaisingh Thangaraj, "Performance Evaluation of Multi-Service Provisioning for Multi-hop Cooperative Data Dissemination in SDHVN ,"***Journal of Ambient Intelligence and Humanized Computing (Q1),*** **(*Springer*)**, DOI:10.1007/s12652-021-03227-4. [**IF: 7.104**].

4. Banoth Ravi, Jaisingh Thangaraj, "Stochastic Traffic Flow Modeling for Multi-hop Cooperative Data Dissemination in VANETs, "***Physical Communication (Q3), (Elsevier)***, DOI: `https://doi.org/10.1016/j.phycom.2021.101290` [**IF: 1.81**].

3. Banoth Ravi, Anmol Gautam, Jaisingh Thangaraj, "Stochastic Performance Modeling and Analysis of Multi Service Provisioning with Software Defined Vehicular Networks,"***AEU-International Journal of Electronics and Communications (Q2), (Elsevier)***, vol. 124, DOI: `https://doi.org/10.1016/j.aeue.2020.153327`, (2020). [**IF: 3.183**].

2. Banoth Ravi, Jaisingh Thangaraj, and Shrinivas Petale, "Data Traffic Forwarding for Inter-vehicular Communication in VANETs Using Stochastic Method, "***Wireless Personal Communications (Q4), (Springer)***, vol. 106, pp. 1591-1607, DOI: `https://doi.org/10.1007/s11277-019-06231-2`, (2019) . [**IF: 1.67**].

1. Jaisingh Thangaraj, Banoth Ravi, and Sunaina Kumari, "Performance analysis of collision avoidance routing protocol for inter-vehicular communication, "***Cluster Computing (Q2), (Springer)***, vol. 22, pp. 7769-7775, DOI: `https://doi.org/10.1007/s10586-017-1381-7`, (2017). [**IF: 1.809**].

Conferences (Published)

5. Banoth Ravi, Jaisingh Thangaraj, "End-to-end delay bound analysis of VANETs based on stochastic method via queueing theory model,"***WiSPNET 2017 (International Conference on Wireless Communications, Signal Processing and Networking), Chennai, India.***,DOI: `10.1109/WiSPNET.2017.8300095`, (2017).

4. Banoth Ravi, Jaisingh Thangaraj, "Stochastic Network Optimization of Data Dissemination for Multi-hop Routing in VANETs,"***WiSPNET 2018 (International Conference on Wireless Communications, Signal Processing and Networking), Chennai, India.***,DOI: `10.1109/WiSPNET.2018.8538353`, (2018).

3. Sailaja, P., Banoth Ravi, and T. Jaisingh, "Performance analysis of AODV and EDAODV routing protocol under congestion control in VANETs,"***2018 Second International Conference on Inventive Communication and Computational Technologies (ICICCT)). IEEE, India.***,DOI: `10.1109/ICICCT.2018.8473050`, (2018).

2. Banoth Ravi, Jaisingh Thangaraj, "Improved Performance Evaluation of Stochastic Modelling and QoS Analysis of Multi-hop Cooperative Data Dissemination in IVC,"***IMICPW 2019 (TEQIP III Sponsored International Conference on Microwave Integrated Circuits, Photonics and Wireless Networks), Tiruchirappalli, India.***,DOI: `10.1109/IMICPW.2019.8933202`, (2019).

1. Banoth Ravi, Anmol Gautam, Thangaraj Jaisingh, Kumar Amitesh, "Improved End to End Delay Bound analysis in Software Defined Mobile Edge Vehicular Networks,"***River Publishers Series in Information Science and Technology), IIT Dhanbad & BIT Sindri, India.***,DOI: `https://doi.org/10.13052/rp-9788770`, (2020).

Technical Skills:

- **Programming Languages:** Java Modelling Tools (JMT), C, MatLab, Python.
- **Frameworks:** EstiNet Network Simulator and Emulator.
- **Industry Software Skills:** Matlab (Int), MS Office, LaTeX.
- **Professional Membership** in IEEE (Member), IEEE Communications Society (ComSoc), OSA, SPIE.

Languages:

Telugu

Hindi

English

Declaration:

I hereby declare that all the details furnished by me are correct to the best of my knowledge and belief.

B. Ravi

(Banoth Ravi)

References:

- **Prof. Jaisingh Thangaraj**, Assistant Professor, Department of ECE, IIT(ISM), Dhanbad,
  `https://iitism.irins.org/profile/97698`
- **Prof. Tarachand Amgoth**, Assistant Professor, Department of CSE, IIT(ISM), Dhanbad,
  `http://iitism.ac.in/~tarachand`
- **Prof. Duggirala Srinivasa Rao**, Professor, Department of ECE, JNTUH College of Engineering, Hyderabad,
  `https://jntuhceh.ac.in/faculty_details/4/dept/342`
- **Prof. Bhukya Mangu**, Professor, Department of EEE, University College of Engineering,Osmania University, Hyderabad,
  `https://eedouce.com/index.php/Faculty_Controller/facultyprofileview/49016`
- **Prof. Sreenu Naik Bhukya**, Assistant Professor, Department of CSE, National Institute of Technology, Calicut
  `http://people.cse.nitc.ac.in/sreenunaikbukya/`